

Continuity and the Intermediate Value Theorem

ANSWER KEY



The three-part continuity test

- 1 State the three conditions that must all hold for f to be continuous at $x = a$.

(1) $f(a)$ is defined; (2) $\lim_{x \rightarrow a} f(x)$ exists; (3) $\lim_{x \rightarrow a} f(x) = f(a)$.

- 2 Determine whether $f(x) = \frac{x^2 - 4}{x - 2}$ (for $x \neq 2$, with $f(2) = 4$ defined separately) is continuous at $x = 2$.

$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x + 2) = 4$. Since $f(2) = 4$ matches, all three conditions hold: f is continuous at $x = 2$.

Finding values that ensure continuity

- 3 Find k so that $f(x) = kx^2$ for $x \leq 2$, and $f(x) = 5x - 2$ for $x > 2$, is continuous at $x = 2$.

Need $4k = 5(2) - 2 = 8$: $k = 2$.

- 4 Find a and b so that $f(x) = 2$ for $x \leq -1$, $f(x) = ax + b$ for $-1 < x < 2$, and $f(x) = 6$ for $x \geq 2$, is continuous everywhere.

At $x = -1$: $-a + b = 2$. At $x = 2$: $2a + b = 6$. Subtracting: $3a = 4$, so $a = \frac{4}{3}$, and $b = 2 + a = \frac{10}{3}$.

Types of discontinuity

- 5 Classify the discontinuity of each function at the indicated point:

a $f(x) = \frac{x+3}{x^2-9}$ at $x = -3$ Removable — $\frac{x+3}{(x-3)(x+3)}$ simplifies to $\frac{1}{x-3}$ for $x \neq -3$.

b $f(x) = \lfloor x \rfloor$ at $x = 2$ Jump — the floor function steps from 1 to 2.

c $f(x) = \frac{1}{(x-1)^2}$ at $x = 1$ Infinite — $f \rightarrow +\infty$ from both sides.

The Intermediate Value Theorem

- 6 Show that $f(x) = x^3 - 3x + 1$ has a root in $(0, 1)$, using the IVT.

$f(0) = 1 > 0$ and $f(1) = 1 - 3 + 1 = -1 < 0$. Since f is continuous (a polynomial) and changes sign on $[0, 1]$, the IVT guarantees a root in $(0, 1)$.

- 7 A continuous function g satisfies $g(1) = -3$ and $g(5) = 7$. Explain why g must take the value 0 somewhere in $[1, 5]$, and state the theorem used.

By the IVT, since g is continuous on $[1, 5]$ and 0 lies between $g(1) = -3$ and $g(5) = 7$, there must exist some $c \in [1, 5]$ with $g(c) = 0$.

- 8 A cup of coffee's temperature (in °C) is a continuous function of time, starting at 90°C and cooling to 25°C over 30 minutes. Explain, using the IVT, why the coffee must have been exactly 50°C at some point.

Temperature is a continuous function of time; since 50 lies between the starting value 90 and ending value 25, the IVT guarantees the temperature equaled 50°C at some instant during the 30 minutes.

Continuity on an interval

- 9 Determine the largest interval(s) on which $f(x) = \sqrt{9 - x^2}$ is continuous.

The expression under the square root must be non-negative: $9 - x^2 \geq 0 \Rightarrow -3 \leq x \leq 3$. f is continuous on the closed interval $[-3, 3]$.

- 10 Determine all values of x where $f(x) = \frac{x+1}{x^2-x-6}$ is discontinuous, and classify each discontinuity.

Factor denominator: $x^2 - x - 6 = (x - 3)(x + 2)$. Discontinuities at $x = 3$ and $x = -2$. Since the numerator $x + 1$ shares no common factor, both are infinite (vertical asymptote) discontinuities, not removable.

Using the IVT to approximate roots

- 11 Show that $f(x) = \cos x - x$ has a root in $(0, 1)$, then use the IVT with the midpoint to narrow the interval to a length-0.5 subinterval.

$f(0) = \cos 0 - 0 = 1 > 0$. $f(1) = \cos 1 - 1 \approx 0.540 - 1 = -0.460 < 0$. Sign change confirms a root exists in $(0, 1)$. Midpoint $x = 0.5$: $f(0.5) = \cos(0.5) - 0.5 \approx 0.878 - 0.5 = 0.378 > 0$. Since $f(0.5) > 0$ and $f(1) < 0$, the root lies in $(0.5, 1)$.

Visualizing a removable discontinuity

- 12 The graph shows $f(x) = \frac{x^2-4}{x-2}$ for $x \neq 2$, simplifying to $f(x) = x + 2$. Use the graph to describe what would need to be true of $f(2)$ for the function to be continuous there.

For continuity at $x = 2$, we would need $f(2)$ to be defined and equal to the limit, $\lim_{x \rightarrow 2} f(x) = 4$. Since the original expression is undefined at $x = 2$ (a hole in the graph at $(2, 4)$), defining $f(2) = 4$ would remove the discontinuity.

Continuity of piecewise and combined functions

- 13 Determine whether $f(x) = |x - 3|$ is continuous at $x = 3$, checking all three conditions.

$f(3) = 0$ is defined. $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (3 - x) = 0$ and $\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (x - 3) = 0$, so $\lim_{x \rightarrow 3} f(x) = 0$. Since $\lim_{x \rightarrow 3} f(x) = 0 = f(3)$, f is continuous at $x = 3$ (even though it has a corner there, which affects differentiability, not continuity).

- 14 If f and g are both continuous at $x = a$, explain why $f(x) + g(x)$ and $f(x) \cdot g(x)$ must also be continuous at $x = a$, citing the relevant limit laws.

By the sum and product limit laws, $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = f(a) + g(a)$, and similarly $\lim_{x \rightarrow a} [f(x)g(x)] = f(a)g(a)$. Since each limit equals the function value at a , both the sum and product are continuous at $x = a$.

An applied IVT problem

- 15 A hiker's elevation (in meters) is a continuous function of distance walked, starting at 200 m and ending at 850 m after 5 km. Explain, using the IVT, why the hiker must have been at exactly 500 m at some point, and state why this reasoning would fail if elevation were not continuous.

Since elevation is continuous over the 5 km hike, and 500 lies between the starting value 200 and ending value 850, the IVT guarantees some distance $c \in [0, 5]$ where elevation equals exactly 500 m. If elevation could jump discontinuously (e.g. teleporting), the function could skip over 500 m entirely without ever equaling it.

Continuity with absolute value and radical functions

- 16 Determine the domain (and hence largest interval of continuity) of $f(x) = \sqrt{x^2 - 4}$.

Need $x^2 - 4 \geq 0$, i.e. $x^2 \geq 4$, so $x \leq -2$ or $x \geq 2$. f is continuous on $(-\infty, -2] \cup [2, \infty)$.

Multiple choice — easy 15

- 17 Find k so that $f(x) = kx$ for $x \leq 2$, and $f(x) = -x - 4$ for $x > 2$, is continuous at $x = 2$.

- a. 3
- b. -3 ✓
- c. -2
- d. -4

Need $k(2) = -1(2) - 4$, so $k = -3$.

18 Find k so that $f(x) = kx$ for $x \leq 2$, and $f(x) = 4x + 2$ for $x > 2$, is continuous at $x = 2$.

a. -5

b. 5 ✓

c. 6

d. 4

Need $k(2) = 4(2) + 2$, so $k = 5$.

19 Find k so that $f(x) = kx$ for $x \leq -2$, and $f(x) = -x + 4$ for $x > -2$, is continuous at $x = -2$.

a. -4

b. -2

c. 3

d. -3 ✓

Need $k(-2) = -1(-2) + 4$, so $k = -3$.

20 Find k so that $f(x) = kx$ for $x \leq -3$, and $f(x) = 4x + 6$ for $x > -3$, is continuous at $x = -3$.

a. 3

b. 1

c. 2 ✓

d. -2

Need $k(-3) = 4(-3) + 6$, so $k = 2$.

21 Find k so that $f(x) = kx$ for $x \leq 1$, and $f(x) = 3x - 3$ for $x > 1$, is continuous at $x = 1$.

- a. 1
- b. 2
- c. 0 ✓
- d. -1

Need $k(1) = 3(1) - 3$, so $k = 0$.

22 Find k so that $f(x) = kx$ for $x \leq 4$, and $f(x) = -3x + 12$ for $x > 4$, is continuous at $x = 4$.

- a. 1
- b. 2
- c. 0 ✓
- d. -1

Need $k(4) = -3(4) + 12$, so $k = 0$.

23 Find k so that $f(x) = kx$ for $x \leq -3$, and $f(x) = -x - 3$ for $x > -3$, is continuous at $x = -3$.

- a. -1
- b. 1
- c. 2
- d. 0 ✓

Need $k(-3) = -1(-3) - 3$, so $k = 0$.

24 Find k so that $f(x) = kx$ for $x \leq 1$, and $f(x) = 0 - 1$ for $x > 1$, is continuous at $x = 1$.

- a. 1
- b. -1 ✓
- c. -2
- d. 0

Need $k(1) = 0(1) - 1$, so $k = -1$.

25 Find k so that $f(x) = kx$ for $x \leq -4$, and $f(x) = -x + 12$ for $x > -4$, is continuous at $x = -4$.

- a. -3
- b. -5
- c. 4
- d. -4 ✓

Need $k(-4) = -1(-4) + 12$, so $k = -4$.

26 Find k so that $f(x) = kx$ for $x \leq -1$, and $f(x) = 0 - 1$ for $x > -1$, is continuous at $x = -1$.

- a. 0
- b. 2
- c. -1
- d. 1 ✓

Need $k(-1) = 0(-1) - 1$, so $k = 1$.

27 Find k so that $f(x) = kx$ for $x \leq 3$, and $f(x) = 2x + 3$ for $x > 3$, is continuous at $x = 3$.

- a. 4
- b. 3 ✓
- c. 2
- d. -3

Need $k(3) = 2(3) + 3$, so $k = 3$.

28 Find k so that $f(x) = kx$ for $x \leq 4$, and $f(x) = -4x + 8$ for $x > 4$, is continuous at $x = 4$.

- a. -1
- b. -3
- c. -2 ✓
- d. 2

Need $k(4) = -4(4) + 8$, so $k = -2$.

29 Find k so that $f(x) = kx$ for $x \leq -1$, and $f(x) = -2x - 3$ for $x > -1$, is continuous at $x = -1$.

- a. -1
- b. 0
- c. 1 ✓
- d. 2

Need $k(-1) = -2(-1) - 3$, so $k = 1$.

30 Find k so that $f(x) = kx$ for $x \leq -4$, and $f(x) = -3x - 8$ for $x > -4$, is continuous at $x = -4$.

- a. 1
- b. -1 ✓
- c. 0
- d. -2

Need $k(-4) = -3(-4) - 8$, so $k = -1$.

31 Find k so that $f(x) = kx$ for $x \leq -4$, and $f(x) = 2x - 12$ for $x > -4$, is continuous at $x = -4$.

- a. 4
- b. -5
- c. 6
- d. 5 ✓

Need $k(-4) = 2(-4) - 12$, so $k = 5$.

Multiple choice — medium 15

32 Find k so that $f(x) = kx^2$ for $x \leq -1$, and $f(x) = 4x - 8$ for $x > -1$, is continuous at $x = -1$.

- a. 12
- b. -11
- c. -12 ✓
- d. -13

Need $k(-1)^2 = 4(-1) - 8 = -12$, so $k = \frac{-12}{1} = -12$.

33 Find k so that $f(x) = kx^2$ for $x \leq 4$, and $f(x) = -2x + 7$ for $x > 4$, is continuous at $x = 4$.

- a. $-\frac{1}{16}$ ✓
- b. $-\frac{17}{16}$
- c. $\frac{15}{16}$
- d. $\frac{1}{16}$

Need $k(4)^2 = -2(4) + 7 = -1$, so $k = \frac{-1}{16} = -\frac{1}{16}$.

34 Find k so that $f(x) = kx^2$ for $x \leq 4$, and $f(x) = 4x - 1$ for $x > 4$, is continuous at $x = 4$.

- a. $-\frac{15}{16}$
- b. $-\frac{1}{16}$
- c. $\frac{31}{16}$
- d. $\frac{15}{16}$ ✓

Need $k(4)^2 = 4(4) - 1 = 15$, so $k = \frac{15}{16} = \frac{15}{16}$.

35 Find k so that $f(x) = kx^2$ for $x \leq -2$, and $f(x) = 5x - 9$ for $x > -2$, is continuous at $x = -2$.

- a. $\frac{19}{4}$
- b. $-\frac{23}{4}$
- c. $-\frac{15}{4}$
- d. $-\frac{19}{4}$ ✓

Need $k(-2)^2 = 5(-2) - 9 = -19$, so $k = \frac{-19}{4} = -\frac{19}{4}$.

36 Find k so that $f(x) = kx^2$ for $x \leq -3$, and $f(x) = -x + 1$ for $x > -3$, is continuous at $x = -3$.

- a. $-\frac{4}{9}$
- b. $-\frac{5}{9}$
- c. $\frac{13}{9}$
- d. $\frac{4}{9}$ ✓

Need $k(-3)^2 = -1(-3) + 1 = 4$, so $k = \frac{4}{9} = \frac{4}{9}$.

37 Find k so that $f(x) = kx^2$ for $x \leq 4$, and $f(x) = -3x + 9$ for $x > 4$, is continuous at $x = 4$.

- a. $-\frac{3}{16}$ ✓
- b. $\frac{3}{16}$
- c. $\frac{13}{16}$
- d. $-\frac{19}{16}$

Need $k(4)^2 = -3(4) + 9 = -3$, so $k = \frac{-3}{16} = -\frac{3}{16}$.

38 Find k so that $f(x) = kx^2$ for $x \leq 1$, and $f(x) = -5x - 1$ for $x > 1$, is continuous at $x = 1$.

- a. -6 ✓
- b. -5
- c. 6
- d. -7

Need $k(1)^2 = -5(1) - 1 = -6$, so $k = \frac{-6}{1} = -6$.

39 Find k so that $f(x) = kx^2$ for $x \leq -4$, and $f(x) = -4x + 3$ for $x > -4$, is continuous at $x = -4$.

a. $\frac{35}{16}$

b. $-\frac{19}{16}$

c. $\frac{19}{16}$ ✓

d. $\frac{3}{16}$

Need $k(-4)^2 = -4(-4) + 3 = 19$, so $k = \frac{19}{16} = \frac{19}{16}$.

40 Find k so that $f(x) = kx^2$ for $x \leq 3$, and $f(x) = -4x - 6$ for $x > 3$, is continuous at $x = 3$.

a. -1

b. -2 ✓

c. -3

d. 2

Need $k(3)^2 = -4(3) - 6 = -18$, so $k = \frac{-18}{9} = -2$.

41 Find k so that $f(x) = kx^2$ for $x \leq 3$, and $f(x) = -4x$ for $x > 3$, is continuous at $x = 3$.

a. $-\frac{7}{3}$

b. $-\frac{1}{3}$

c. $\frac{4}{3}$

d. $-\frac{4}{3}$ ✓

Need $k(3)^2 = -4(3) = -12$, so $k = \frac{-12}{9} = -\frac{4}{3}$.

42 Find k so that $f(x) = kx^2$ for $x \leq 3$, and $f(x) = -x$ for $x > 3$, is continuous at $x = 3$.

- a. $-\frac{4}{3}$
- b. $\frac{1}{3}$
- c. $\frac{2}{3}$
- d. $-\frac{1}{3}$ ✓

Need $k(3)^2 = -1(3) = -3$, so $k = \frac{-3}{9} = -\frac{1}{3}$.

43 Find k so that $f(x) = kx^2$ for $x \leq -4$, and $f(x) = -4x - 8$ for $x > -4$, is continuous at $x = -4$.

- a. $\frac{1}{2}$ ✓
- b. $-\frac{1}{2}$
- c. $\frac{3}{2}$
- d. -2

Need $k(-4)^2 = -4(-4) - 8 = 8$, so $k = \frac{8}{16} = \frac{1}{2}$.

44 Find k so that $f(x) = kx^2$ for $x \leq -1$, and $f(x) = 4x + 6$ for $x > -1$, is continuous at $x = -1$.

- a. -2
- b. 1
- c. 2 ✓
- d. 3

Need $k(-1)^2 = 4(-1) + 6 = 2$, so $k = \frac{2}{1} = 2$.

45 Find k so that $f(x) = kx^2$ for $x \leq 2$, and $f(x) = -x$ for $x > 2$, is continuous at $x = 2$.

- a. -1
- b. $-\frac{3}{2}$
- c. $-\frac{1}{2}$ ✓
- d. $\frac{1}{2}$

Need $k(2)^2 = -1(2) = -2$, so $k = \frac{-2}{4} = -\frac{1}{2}$.

46 Find k so that $f(x) = kx^2$ for $x \leq 3$, and $f(x) = x - 3$ for $x > 3$, is continuous at $x = 3$.

- a. 2
- b. -1
- c. 0 ✓
- d. 1

Need $k(3)^2 = 1(3) - 3 = 0$, so $k = \frac{0}{9} = 0$.

Multiple choice — hard 15

47 Find a so that $f(x) = 0$ for $x \leq 0$, $f(x) = ax + b$ for $0 < x < 3$, and $f(x) = -2$ for $x \geq 3$, is continuous everywhere.

- a. $-\frac{2}{3}$ ✓
- b. $\frac{1}{3}$
- c. $-\frac{5}{3}$
- d. $\frac{2}{3}$

At $x = 0$: $a(0) + b = 0$. At $x = 3$: $a(3) + b = -2$. Subtracting: $a(3 - (0)) = -2 - (0)$, so $a = \frac{-2}{3} = -\frac{2}{3}$.

48 Find a so that $f(x) = 1$ for $x \leq -5$, $f(x) = ax + b$ for $-5 < x < 3$, and $f(x) = 4$ for $x \geq 3$, is continuous everywhere.

- a. $-\frac{3}{8}$
- b. $\frac{3}{8}$ ✓
- c. $-\frac{5}{8}$
- d. $\frac{11}{8}$

At $x = -5$: $a(-5) + b = 1$. At $x = 3$: $a(3) + b = 4$. Subtracting: $a(3 - (-5)) = 4 - (1)$, so $a = \frac{3}{8} = \frac{3}{8}$.

- 49 Find a so that $f(x) = 3$ for $x \leq -3$, $f(x) = ax + b$ for $-3 < x < 5$, and $f(x) = 4$ for $x \geq 5$, is continuous everywhere.

- a. $\frac{9}{8}$
- b. $-\frac{7}{8}$
- c. $-\frac{1}{8}$
- d. $\frac{1}{8}$ ✓

At $x = -3$: $a(-3) + b = 3$. At $x = 5$: $a(5) + b = 4$. Subtracting: $a(5 - (-3)) = 4 - (3)$, so $a = \frac{1}{8} = \frac{1}{8}$.

- 50 Find a so that $f(x) = -5$ for $x \leq -3$, $f(x) = ax + b$ for $-3 < x < 5$, and $f(x) = -4$ for $x \geq 5$, is continuous everywhere.

- a. $-\frac{7}{8}$
- b. $-\frac{1}{8}$
- c. $\frac{9}{8}$
- d. $\frac{1}{8}$ ✓

At $x = -3$: $a(-3) + b = -5$. At $x = 5$: $a(5) + b = -4$. Subtracting: $a(5 - (-3)) = -4 - (-5)$, so $a = \frac{1}{8} = \frac{1}{8}$.

- 51 Find a so that $f(x) = 3$ for $x \leq -1$, $f(x) = ax + b$ for $-1 < x < 3$, and $f(x) = 5$ for $x \geq 3$, is continuous everywhere.

- a. $\frac{3}{2}$
- b. $\frac{5}{2}$
- c. $-\frac{1}{2}$
- d. $\frac{1}{2}$ ✓

At $x = -1$: $a(-1) + b = 3$. At $x = 3$: $a(3) + b = 5$. Subtracting: $a(3 - (-1)) = 5 - (3)$, so $a = \frac{2}{4} = \frac{1}{2}$.

- 52 Find a so that $f(x) = -6$ for $x \leq -1$, $f(x) = ax + b$ for $-1 < x < 1$, and $f(x) = -3$ for $x \geq 1$, is continuous everywhere.

- a. $\frac{1}{2}$
- b. $\frac{5}{2}$
- c. $-\frac{3}{2}$
- d. $\frac{3}{2}$ ✓

At $x = -1$: $a(-1) + b = -6$. At $x = 1$: $a(1) + b = -3$. Subtracting: $a(1 - (-1)) = -3 - (-6)$, so $a = \frac{3}{2} = \frac{3}{2}$.

- 53 Find a so that $f(x) = -6$ for $x \leq -2$, $f(x) = ax + b$ for $-2 < x < 1$, and $f(x) = 6$ for $x \geq 1$, is continuous everywhere.

- a. 4 ✓
- b. 5
- c. 3
- d. -4

At $x = -2$: $a(-2) + b = -6$. At $x = 1$: $a(1) + b = 6$. Subtracting: $a(1 - (-2)) = 6 - (-6)$, so $a = \frac{12}{3} = 4$.

- 54 Find a so that $f(x) = -3$ for $x \leq -4$, $f(x) = ax + b$ for $-4 < x < 5$, and $f(x) = 3$ for $x \geq 5$, is continuous everywhere.

- a. $\frac{2}{3}$ ✓
- b. $-\frac{2}{3}$
- c. $-\frac{1}{3}$
- d. $\frac{5}{3}$

At $x = -4$: $a(-4) + b = -3$. At $x = 5$: $a(5) + b = 3$. Subtracting: $a(5 - (-4)) = 3 - (-3)$, so $a = \frac{6}{9} = \frac{2}{3}$.

- 55 Find a so that $f(x) = 1$ for $x \leq -4$, $f(x) = ax + b$ for $-4 < x < 1$, and $f(x) = 4$ for $x \geq 1$, is continuous everywhere.

a. $\frac{3}{5}$ ✓

b. $-\frac{2}{5}$

c. $\frac{8}{5}$

d. $-\frac{3}{5}$

At $x = -4$: $a(-4) + b = 1$. At $x = 1$: $a(1) + b = 4$. Subtracting: $a(1 - (-4)) = 4 - (1)$, so $a = \frac{3}{5} = \frac{3}{5}$.

- 56 Find a so that $f(x) = 2$ for $x \leq 0$, $f(x) = ax + b$ for $0 < x < 1$, and $f(x) = 4$ for $x \geq 1$, is continuous everywhere.

a. 2 ✓

b. 1

c. -2

d. 3

At $x = 0$: $a(0) + b = 2$. At $x = 1$: $a(1) + b = 4$. Subtracting: $a(1 - (0)) = 4 - (2)$, so $a = \frac{2}{1} = 2$.

- 57 For $f(x) = x^3 - 4x + 1$, evaluate $f(2)$ and $f(3)$, and state whether the IVT guarantees a root in $(2, 3)$.

a. $f(2) = 1$, $f(3) = 16$ — same sign, but a root is still guaranteed since f is a polynomial.

b. $f(2) = 1$, $f(3) = 16$ — same sign, so the IVT does not guarantee a root in $(2, 3)$. ✓

c. $f(2) = 1$, $f(3) = 16$ — the IVT cannot be applied since f is not continuous.

d. $f(2) = 1$, $f(3) = 16$ — opposite signs, so the IVT guarantees a root in $(2, 3)$.

$f(x)$ is a polynomial, hence continuous everywhere. $f(2) = 1$, $f(3) = 16$ — same sign, so the IVT does not guarantee a root in $(2, 3)$.

58 For $f(x) = x^3 - 4x + 5$, evaluate $f(-3)$ and $f(-2)$, and state whether the IVT guarantees a root in $(-3, -2)$.

- a. $f(-3) = -10$, $f(-2) = 5$ — the IVT only applies to linear functions, not cubics.
- b. $f(-3) = -10$, $f(-2) = 5$ — same sign, so the IVT guarantees no root in $(-3, -2)$.
- c. $f(-3) = -10$, $f(-2) = 5$ — **opposite signs, so the IVT guarantees a root in $(-3, -2)$. ✓**
- d. $f(-3) = -10$, $f(-2) = 5$ — opposite signs, but the IVT cannot be applied since f is not continuous.

$f(x)$ is a polynomial, hence continuous everywhere. $f(-3) = -10$, $f(-2) = 5$ — opposite signs, so the IVT guarantees a root in $(-3, -2)$.

59 For $f(x) = x^3 - 3x + 6$, evaluate $f(-1)$ and $f(0)$, and state whether the IVT guarantees a root in $(-1, 0)$.

- a. $f(-1) = 8$, $f(0) = 6$ — opposite signs, so the IVT guarantees a root in $(-1, 0)$.
- b. $f(-1) = 8$, $f(0) = 6$ — same sign, but a root is still guaranteed since f is a polynomial.
- c. $f(-1) = 8$, $f(0) = 6$ — **same sign, so the IVT does not guarantee a root in $(-1, 0)$. ✓**
- d. $f(-1) = 8$, $f(0) = 6$ — the IVT cannot be applied since f is not continuous.

$f(x)$ is a polynomial, hence continuous everywhere. $f(-1) = 8$, $f(0) = 6$ — same sign, so the IVT does not guarantee a root in $(-1, 0)$.

60 For $f(x) = x^3 - 6x - 1$, evaluate $f(0)$ and $f(1)$, and state whether the IVT guarantees a root in $(0, 1)$.

- a. $f(0) = -1$, $f(1) = -6$ — opposite signs, so the IVT guarantees a root in $(0, 1)$.
- b. $f(0) = -1$, $f(1) = -6$ — same sign, but a root is still guaranteed since f is a polynomial.
- c. $f(0) = -1$, $f(1) = -6$ — the IVT cannot be applied since f is not continuous.
- d. $f(0) = -1$, $f(1) = -6$ — **same sign, so the IVT does not guarantee a root in $(0, 1)$. ✓**

$f(x)$ is a polynomial, hence continuous everywhere. $f(0) = -1$, $f(1) = -6$ — same sign, so the IVT does not guarantee a root in $(0, 1)$.

61 For $f(x) = x^3 + 6x + 1$, evaluate $f(-1)$ and $f(0)$, and state whether the IVT guarantees a root in $(-1, 0)$.

a. $f(-1) = -6$, $f(0) = 1$ — opposite signs, but the IVT cannot be applied since f is not continuous.

b. $f(-1) = -6$, $f(0) = 1$ — same sign, so the IVT guarantees no root in $(-1, 0)$.

c. $f(-1) = -6$, $f(0) = 1$ — the IVT only applies to linear functions, not cubics.

d. $f(-1) = -6$, $f(0) = 1$ — **opposite signs, so the IVT guarantees a root in $(-1, 0)$.** ✓

$f(x)$ is a polynomial, hence continuous everywhere. $f(-1) = -6$, $f(0) = 1$ — opposite signs, so the IVT guarantees a root in $(-1, 0)$.